

CSC-355 Homework #3

context free language, (E)BNF, and parsing

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This homework is an opportunity for you to practice building parse trees, given an (E)BNF that describes a context free language.

Problem 1: BNF Generation

Generate BNF grammars for each of the following language descriptions.

- (a) A language for all strings that look like n repeated **a**'s followed by a single **b** followed by n **a**'s.
- (b) A language for all strings that look like $2n$ **a**'s followed by n **b**'s.
- (c) A language for all strings that look like $3n$ **a**'s.

Problem 2: BNF Parsing

Demonstrate the right-most derivation for each of the strings below (i.e., always replace the right-most non-terminal when deriving the string). For one string in each of the three following languages, draw out the parse tree. Make sure your parse trees contain the complete information from the parse. Ignore spaces.

L_1 :

$S \rightarrow T \mid S \text{ "+" } T$

$T \rightarrow F \mid T \text{ "*" } F$

$F \rightarrow \text{"0"} \mid \text{"1"} \mid \text{"2"} \mid \text{"3"} \mid \text{"4"}$

L_2 :

$S \rightarrow M V \mid M V O$

$M \rightarrow \text{"the"} A N \mid \text{"the"} N$

$A \rightarrow \text{"intelligent"} \mid \text{"smart"} \mid \text{"lazy"}$

$N \rightarrow \text{"person"} \mid \text{"robot"}$

$V \rightarrow \text{"destroys"} \mid \text{"creates"}$

$O \rightarrow \text{"the cake"} \mid \text{"the house"} \mid \text{"the program"}$

L_3 :

$S \rightarrow V \mid \text{"\lambda"} V \text{"." } S \mid S S \mid (S)$

$V \rightarrow \text{"w"} \mid \text{"x"} \mid \text{"y"} \mid \text{"z"}$

- (a) (use L_1): $2 + 2$
- (b) (use L_1): $3 + 3 + 3$
- (c) (use L_1): $2 + 2 * 3 * 3$
- (d) (use L_2): the lazy person destroys

- (e) (use L_2): the intelligent robot creates the cake
- (f) (use L_2): the smart person creates the program
- (g) (use L_3): $\lambda x.x$
- (h) (use L_3): $wxyz$
- (i) (use L_3): $(\lambda w.w)(\lambda z.z)$
- (j) Give a string that demonstrates the ambiguity of L_3 .

Problem 3: EBNF

- (a) What is the difference between square brackets ($[]$) and curly braces ($\{\}$)?
- (b) What is the difference between curly braces ($\{\}$) and parentheses $()$?
- (c) Describe the following production rule in words: $S \rightarrow \text{"f"} \{ (a \mid b) \}$
- (d) Describe the following production rule in words: $S \rightarrow \text{"f"} [\text{"g"}] \text{"h"}$